

Infix to prefix algorithm

27 April 2023 11:58

1) start scanning from **right to left**

Infix expression:

2) If(current element is an operand)

$5*3/2*6/4+7-3+8/2$

Stack :

Append it to the prefix expression

Else

// If(current element is an operator)

Current element :

While(stack is not empty && priority of topmost element > priority of current element)

{

Pop element from the stack and append it to prefix expression

Prefix Expression: $28/3746235*/*/+--$

Reverse: $+-+/*/*5326473/82$

}

Push current element onto the stack.

3) Repeat the above steps till end of infix expression

4) Pop all the remaining elements from the stack one by one and append them to prefix expression

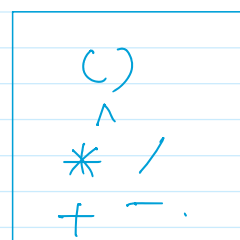
5) **Reverse the prefix expression**

If the current element is the closing bracket,

Push it on to the stack

If the current element is the opening bracket,

- Pop elements from the stack and append them to prefix exp till its corresponding closing brackets does not occur
- Pop opening brackets from the stack and discard both the brackets



$5*3/2*6/4+7-3+8/2$

① ② ③ ④ ⑤ ⑥ ⑦ ⑧ ⑨

$5 * 3 / 2 * 6 / 4 + 7 - 3 + 8 / 2$

$* 5 3 / 2 * 6 / 4 + 7 - 3 + 8 / 2$

$/ * 5 3 2 * 6 / 4 + 7 - 3 + 8 / 2$

$* / * 5 3 2 6 / 4 + 7 - 3 + 8 / 2$

$/ * / * 5 3 2 6 4 + 7 - 3 + 8 / 2$

$/ * / * 5 3 2 6 4 + 7 - 3 + / 8 2$

$+ / * / * 5 3 2 6 4 7 - 3 + / 8 2$

$- + / * / * 5 3 2 6 4 7 3 + / 8 2$

$$\underbrace{- + / * / * 5326473}_{\downarrow} + \underline{182}$$

$+ - + / * / * 5326473 / 82$

+-+/*/*5326473/82